Nathan Alvares

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.14
Aditya English Medium School	Class XII (CBSE)	2024	94.20%
PICT Model School	Class X (CBSE)	2022	97.80%

WORK EXPERIENCE

Undergraduate Research Intern | Transportation Systems Engineering Lab, IIT Hyderabad

Supervisor: Prof. Digvijay Pawar

Developed and curated road and traffic sign datasets for road safety applications, supporting computer vision model training and intelligent transportation system research.

PROJECTS

Automated Fest Pass Distribution System | Elan & nVision 2026

- **Architected and deployed** a production-grade **Node.js** backend service on a **Microsoft Azure Virtual Machine** that automated the complete fest pass distribution pipeline by processing participant registrations from **Google Forms/Google Sheets**, eliminating manual intervention for **21,000-25,000** registrations.
- **Designed a fault-tolerant background worker** that periodically polled registration data, generated personalized **Code128** barcode passes using **bwip-js**, rendered dynamic HTML email templates with **Handlebars**, and delivered customized passes through **Gmail SMTP** using **Nodemailer**.
- **Engineered production reliability features** including persistent checkpointing for crash recovery, row-level processing locks to prevent duplicate processing, SMTP connection pooling, buffered batch synchronization, automatic sender-account failover for Gmail quota exhaustion, and idempotent recovery mechanisms for transient failures.
- **Optimized throughput and operational efficiency** by processing registrations incrementally, batching Google Sheets API updates, reusing SMTP connections, and maintaining persistent worker state, enabling reliable large-scale pass generation with minimal operational overhead during the three-day festival.

Tech Stack: Node.js, JavaScript, Microsoft Azure VM, Google Sheets API, Nodemailer, Handlebars, bwip-js

Premier League Match Outcome Predictor

- **Developed an end-to-end machine learning pipeline** to predict **English Premier League** match outcomes (Home Win / Draw / Away Win) using historical match data spanning multiple seasons, framing the problem as a **multi-class classification** task using **Random Forest**.
- **Designed a comprehensive data preprocessing pipeline** using **Pandas** and **NumPy**, including missing value handling, duplicate removal, categorical encoding, temporal feature extraction, rolling-window feature generation, and dataset preparation while preventing data leakage through historical feature construction.
- **Engineered predictive features** from multiple dimensions, including **match statistics** (GF, GA, xG, xGA, possession, shots, shots on target, free kicks, penalty kicks), **contextual information** (venue, opponent, formation, kickoff hour, matchday), and **rolling historical performance metrics** to capture recent team form, attacking efficiency, defensive stability, and home-field advantage.
- **Trained and optimized a Random Forest Classifier** to model complex non-linear relationships between engineered features and match outcomes, leveraging ensemble learning to improve generalization and reduce overfitting on structured football data.
- **Evaluated model performance** using **accuracy, precision, recall, F1-score, confusion matrix, and feature importance analysis**, enabling iterative feature refinement and identification of the statistical factors contributing most significantly to match outcome prediction.
- **Built a modular and reproducible Python pipeline** separating data ingestion, preprocessing, feature engineering, model training, prediction, and evaluation stages, allowing easy retraining on future seasons and extensibility for experimenting with alternative machine learning algorithms.

Tech Stack: Python, Pandas, NumPy, scikit-learn, Random Forest, Data Analysis

Custom Unix Shell in C++

- Built a Unix-like command-line shell in modern C++ using POSIX system calls (fork, execvp, waitpid, dup2) and terminal APIs (termios, signals).
- Implemented custom command parsing with support for multi-stage pipelines, input/output redirection (<, >, >>), background process execution (&), and built-in shell commands.
- Developed a custom interactive line editor in raw terminal mode featuring command history navigation, cursor movement, in-line insertion/deletion, and real-time prompt redrawing.
- Designed process management and job tracking infrastructure, including background job monitoring and asynchronous child-process reaping.
- Added signal handling for interactive terminal control and engineered the shell using a modular architecture with separate parser, executor, terminal, history, and shell-management components.

Tech Stack: C++, POSIX APIs (fork, execvp, waitpid, dup2, pipe), Linux System Programming, Unix Signals, termios, STL, Makefile, Git

SKILLS

Languages: C, C++, Python, JavaScript, TypeScript, SQL, HTML, CSS

Web Development: React.js, Next.js, FastAPI, REST APIs

Databases: PostgreSQL, SQLite

Machine Learning: Scikit-learn, Pandas, NumPy, Seaborn

Tools: Git, GitHub, Docker, Jupyter Notebook, Postman

Core CS: DSA, OOP, DBMS

RELEVANT COURSES

Mathematics: Calculus, Probability, Statistics, Linear Algebra, Transform Techniques, Differential Equations

POSITIONS OF RESPONSIBILITY

Dev Core at Lambda, The Software Development Club at IIT Hyderabad

Web Coordinator at Elan & nVision, the annual Tech and Cultural Fest of IIT Hyderabad